

Lesson 4-3 · Period 2

Multiply & Divide Rational Expressions +
DOK-3 Closure

Klimsara-adapted · Student-centered · No Blooket

Do Now — Simplify and state the domain

Algebra

2 · Lesson 4-3 · Period 2

DOK 1

5 min

bridge from P1

Do Now · Simplify this rational expression What is the simplified form of the rational expression? What is the domain?

$$\frac{x^3 + 9x^2 - 10x}{x^3 - 9x^2 - 10x}$$

Factor every numerator and denominator completely. State all excluded values.

Launch — Example 3 (Multiply & Divide Rules)

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DOK 2

12 min

teacher models

Multiply / Divide Rules (keep on your page)

1. Factor every numerator AND denominator completely.
2. For division, rewrite as multiplication by the **RECIPROCAL** of the divisor.
3. Cancel only common **FACTORS** across any numerator with any denominator.
4. State the domain: exclude every zero of every denominator that appeared in **ANY** step, including the reciprocal's new denominator.

Explore — TPS + DOK-3 Performance Task

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DOK 2-3

33 min

student work

7 items in order. Plan ~12 min for Practice #13 (Closure Argument).

Try It 3 · Multiply rational expressions Factor, cancel, state domain. (Two parts.)

Practice #27 · Multiply with factoring $\frac{(x - y)^2}{x + y}$.

$\frac{3x + 3y}{x^2 - y^2}$ — find product and domain.

DOK 2

5 min

academic conversation + P3 bridge

Sentence frame · Closure Argument “I rewrite the division as multiplication by the reciprocal, giving _____. After factoring, I cancel the factor _____, leaving _____. The domain excludes $x =$ _____ because those values make at least one denominator zero at some step.”

Bridge to Period 3 Next period we apply multiply/divide skills to complex rational expressions and assessment critical

Exit Ticket — Summary Recap

Algebra 2 · Lesson 4-3 ·

Period 2

DOK 1-2

3 min

not graded

Complete on your exit ticket

1. To divide rational expressions I first _____.
2. I cancel common factors, not _____.
3. My domain restriction list must include zeros from _____.
4. One biggest thing I learned today: _____.